

Prediction of Wax Disappearance Temperature by Intelligent Models

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ABSTRACT: It is well-known that reservoir hydrocarbon fluids contain heavy paraffins that may form solid phases of wax at low temperatures. Problems associated with wax formation and deposition are a major concern in production and transportation of hydrocarbon fluids. Thus, testing of wax disappearance temperature (WDT) is essential in high-efficiency development of crude oil. For the sake of reduction of time and improvement of accuracy, four metaheuristic models called gray wolf optimizer-based support vector machine (GWO-SVM), least-squares support vector machine, genetic algorithm-based adaptive network-based fuzzy inference system, and particle swarm optimization-based adaptive network-based fuzzy inference system were used for the prediction of WDT in binary, ternary, and multicomponent systems in the range of 0.1–100 MPa. The input parameters are molar mass and pressure, and the output is the WDT at every point. The comparison between the four models shows that the GWO-SVM gets the best accordance with experimental data sets with the minimum average absolute relative deviation (AARD = 0.7128%), maximum determination coefficient ($R^2 = 0.9546$), and minimum root-mean-squared error (RMSE = 2.4208) in all 272 data points. And outliers detection using the leverage approach to detect the doubt points, where only 6 data points in all 272 data points.

1. INTRODUCTION

Crude oil and gas condensate are made up of a large number of long *n*-alkane chains. In the event when the temperature drops to a temperature below the wax appearance temperature (WAT), there is wax precipitation in the transport equipment and pipelines, which may lead to plugging of well bores and blockage of pipeline. To address the issue of wax deposition, we need to know how it appeared.

The temperature reduction severely affects the precipitation of wax, stemmed from gas expansion and heat losses from oil and gas streams to the environments. Besides, wax precipitation is also resulted from segregation of dissolved gases and evaporation of the light fraction of the oil mixtures.¹ The main component of petroleum waxes is normal alkanes.² Therefore, many research studies preceed with these constituents to get the accurate knowledge of wax precipitation.

When wax is first perceived at cooling in a given pressure, the temperature is cloud point temperature, that is, wax appearance temperature (WAT), mostly measured in laboratories. However, WAT is not certainly an equilibrium point, which lies on measurements. Experimental studies show that WAT is commonly related to the cooling rate extremely; the faster the cooling rate, the lower the measured WAT. For example, the measured WAT obtained using visual microscopy is 10–20 °C higher than that obtained using “laser-based solid detection systems”, “differential scanning calorimetry”, and “viscometry”.^{3,4}

Different from WAT, the wax disappearance temperature (WDT) stands for a certain solid–liquid equilibrium point. Advisable experimental techniques and methods (i.e., equilibrium step heating) make a great WDT's value, where measured WDTs are in agreement with the standard values within a reasonable error range in different laboratories.

There are four approaches to determine the accurate WDT, which are experimental measurements, thermodynamic models, empirical models, and intelligent algorithms.

The experimental measurement is the most reliable one of the four approaches. Some researchers conducted experiments to determine WDT. Metivaud (1999)⁵ et al. measured WAT of consecutive C14 – C21 combining differential scanning calorimetry and X-ray analyses, roughly covering from 0 to 50 °C only at atmosphere pressure. There is a great limitation of pressure and temperature. To overcome the nonideality of solution to overestimate the precipitation, Dauphin et al. (1999)⁶ measured the WAT, waxy solid content, and wax composition of a wax system ranging from C18 to C36 at atmosphere pressure, but some paraffins were removed from the series so that it can be provided with various bimodal distributions, covering from 300 to 315 K, which leads to incompleteness of the tested system and inaccuracy of the experiment. However, it is assumed that there is coexistence of several solid solutions in systems. To extend the pressure range, Daridon et al. (2002)⁷ used a high-pressure apparatus resting on a polarizing microscope, which is able to determine the visual liquid–solid phase transition in complex systems and measure the WDT at a wider range of pressures below 100 MPa, testing a mixture of *n*-paraffins ranging from C13 to C24. However, the experiment setup is so expensive that it cannot not be popularized. Milhet (2005)⁸ measured WDT of two binary systems, C14 + C15 and C14 + C16, under high pressure using a polar microscope. However, the data received are higher than the data from Metivaud (1999)⁵ for the C14 + C16 system. The reason may be appearance of a triclinic phase with increasing pressure,

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which cannot be measured any longer. To overcome the drawback, Rizzo (2007)⁹ designed an apparatus based on two sapphire windows and a high-pressure cell coupled with a detection system of the reflected and refracted light intensities coming from a laser beam. This was designed to observe the phase transformation. Although the experimental value of WDT is usually available at increasing pressures, it is expensive, laborious, and time-consuming.¹⁰

There is a complex calculation and realization of critical parameters of wax in thermodynamic models, and it is necessary to understand mathematical principles and analytical solution for establishing equations and computing the numerical solution. Won (1986)¹¹ and Won (1989)¹² presented two modified regular solutions for wax deposition. Hansen et al. (1998) presented an improved regular solution employing Flory's theory (1953)¹³ of multicomponent solutions for activity coefficient. Besides, Pedersen et al. (1991)¹⁴ developed the Won (1986)¹¹ modified model for WAT calculations. Lira-Galeana et al. (1996)¹⁵ developed a multi-solid phase for wax precipitation and regarded every solid phase as a pure component, considering the ideality only. Therefore, Continho (1998)¹⁶ improved a predictive UNIQUAC model for solid and liquid phase equilibria considering the nonideality based on the model proposed by Abrams and Prausnitz (1975),¹⁷ which allows solid splits, instead of Wilson, although it describes the liquid phase inaccurately. For considering the Poynting correction, Zuo et al. (2001)¹⁸ raised a solid-solution model to calculate wax precipitation in crude oils and gas condensates, but it overestimates the WAT and amount of precipitated wax. In the study about the investigation of the phase behaviors between the wax disappearance temperature (WDT) and the wax appearance temperature (WAT) at atmosphere pressure, Bhat and Mehrotra (2004)¹⁹ found that the WDT is closer to the saturation temperature of wax precipitation, not WAT. For exploring the effect of pressure (between 0.1 and 50 MPa) on wax phase equilibria, Ji et al. (2004)²⁰ managed to calculate WDT for binary and multicomponent systems by applying the UNIQUAC thermodynamic model. Ghanaei et al. (2007)²¹ developed a new improved model for the prediction of composition of wax and WDT of binary systems of C14 – C15 and C14 – C16 (0.1–100 MPa). Nasrifar and Kheshty (2011)²² investigated a new method about the estimation of the Poynting correction term to broaden the pressure range and managed a model for the prediction at low and high pressures of wax precipitation. Ghanaei (2012)²³ presented a model applying heat capacity correlations to explore the solid–solid phase transition appearing in solid–liquid fugacity of a pure component as phase change and heat enthalpy, thus describing nonideality of the solid wax phase. Yang et al. (2016)²⁴ improved the Wilson model considering the molecules' difference and entropy contribution only and developed a new solid-solution model for liquid phase with regular solution theory and Flory free-volume theory and for solid phase with solid-solution theory, considering nonideality of the solid phase, enthalpy contribution, the energetic interactions between the components, entropy contribution, and differences between the sizes and shapes of the molecules.

Various empirical studies such as those of Pauly et al., 1998;²⁵ Metivaud et al., 1999;⁵ Pauly et al., 2001;^{7,26} Daridon et al., 2002;⁷ Pauly et al., 2004;²⁷ Milhet et al., 2005;⁸ Rizzo et al., 2007;⁹ and Vafaie-Sefti et al., 2007²⁸ were accomplished on wax precipitation.

The intelligent algorithm has been developed rapidly and implemented to link WDT and different parameters. ANNs as

utilizable methods are beneficial for modeling complicated systems in petroleum engineering areas such as prediction of WDT. Moradi (2013)²⁹ et al. proposed ANN, in which a tangs transfer function is used in three-layer feedforward backpropagation, in predicting WDT at elevated molar mass (87.82–293.90) and pressure from 0.1–100.3 MPa with agreeable results. For many years, the modelers either had to manually tune the parameters during a time-consuming trial-and-error process or used the default settings for simplicity, which are usually far from optimal. Metaheuristic optimization algorithms have recently emerged as a remedy to alleviate the difficulty associated with the machine learning methods. Arash Kamari (2013)³⁰ developed a predictive model for calculation of wax precipitation, least-squares support vector machine (LSSVM) modeling combined with the coupled simulated annealing (CSA) approach, achieving a great agreement between predictions based on the CSA-LSSVM and experimental data on wax precipitation. With the increasing progress of intelligence algorithms for many years, a gray wolf optimizer (GWO) has been reliable for settling numerous optimization issues, like nonconvex economic load dispatch.³¹ Menad (2018)³² proposed radial basis function neural network-GWO for modeling temperature-based oil–water relative permeability. Zhou (2018)³³ proposed differential evolution-SVM to establish the color difference classification model of printing and dyeing, for its good global searching ability. Jaafari (2019)³⁴ developed adaptive network-based fuzzy inference system (ANFIS)-GWO to estimate landslide susceptibility, for which GWO is able to overcome the shortcoming of the ANFIS model and optimize base parameters. However, it is not very common that researchers utilize the intelligent algorithm for working out the WDT in the literature. Besides the application of particle swarm optimization (PSO) and genetic algorithm (GA) to optimize the control parameters of SVM, in this work, a new framework is provided to optimize these parameters by implementing a new recent metaheuristic algorithm (GWO) that is shown to be more efficient in the optimization process compared with the well-known metaheuristic algorithms.

The target in the work is to propose and verify four metaheuristic models on the basis of reliable experimental WDT data from the existing literature. In this work, four models were proposed for prediction of the WDT. The input parameters of the proposed models are the pressure and molar mass, and the objective parameter is WDT. The accuracy and reliability of the four proposed models were examined by applying 272 data sets available in the literature.

This work was organized as follows:

- In part 2, the brief introduction of GWO-SVM, LSSVM, GA-ANFIS, and PSO-ANFIS models was given.
- In part 3, the used data and factors affecting WDT were given.
- In part 4, comparing the results of models with experimental data, the performance of the four metaheuristic models was validated.
- In part 5, the corresponding conclusions were given.

2. METHODOLOGY

2.1. GWO-SVM. **2.1.1. Support Vector Machine.** SVM³⁵ is a statistical learning theory regression method, whose main philosophy is to draw the input data (low-dimensional space) into a high-dimensional feature space and then to build a kernel function for getting a linear regression function. The input data sets are as follows: $\{(x_{i(m)}, y_i), i = 1, 2, \dots, N; m = 1, 2, \dots, n\}$, where

$x_i^{(m)}$ denotes the i th element in the m th-dimensional space and y_i and N are the scalar output and the aggregate of data sets, respectively.

The support vector regression function is as follows eq 1,1

$$f(x) = \langle \lambda, x \rangle + b \quad (1)$$

where λ is the weight vector and b is the threshold value of the regression function, which is the deviation term.

According to the minimizing regularized risk function as follows, the regression issue is settled

$$R_{\text{emp}}(f) = \frac{1}{N} \sum_{i=1}^N L_{\varepsilon}[y_i - f(\lambda, x_i)] + \frac{1}{2} \|\lambda\|^2 \quad (2)$$

where

$$L_{\varepsilon}[y_i - f(\lambda, x_i)] = \begin{cases} 0, & \text{if } |y - f(\lambda, x_i)| \leq \varepsilon \\ |y - f(\lambda, x_i)| - \varepsilon, & \text{otherwise} \end{cases} \quad (3)$$

Equation 3 gives the ε -insensitive loss function developed by Vapnik (1995),³⁶ and the range of the regression function $f(x)$ is limited by the ε -value

However, the error value between the experimental and estimated values is within the range of ε -value, and the loss value is regarded as zero. If not, the loss is treated as the absolute difference between the predicted value and the ε -value. The optimization object is expressed by

$$\text{minimize } \frac{1}{2} \|\lambda\|^2 + c \sum_{i=1}^N (\xi + \xi^*) \quad (4)$$

$$\text{subject to } \begin{cases} y_i - f(x_i, \lambda) - b \leq \varepsilon + \xi^* \\ f(\lambda, x_i) + b - y_i \leq \varepsilon + \xi^* \\ \xi_i^*, \xi_i \geq 0 \end{cases} \quad (5)$$

where the regularization parameter C is the penalty term. The slack variables ξ and ξ^* served as bias restriction of training data sets beyond the range of ε -insensitive. This optimization issue could be transformed into a dual issue. The formula is converted into the following function with the optimization theory

$$\begin{aligned} \text{maximize } & -\frac{1}{2} \sum_{i,j=1}^n (\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*)(x_i, x_j) \\ & - \varepsilon \sum_{i=1}^n (\alpha_i + \alpha_i^*) + \sum_{i=1}^n y_i (\alpha_i - \alpha_i^*) \end{aligned} \quad (6)$$

$$\text{subject to } \begin{cases} \sum_{i=1}^n (\alpha_i - \alpha_i^*) = 0 \\ 0 \leq \alpha_i, \alpha_i^* \leq C \end{cases} \quad (7)$$

where α_i and α_i^* are, respectively, the Lagrange multiplier and the optimal desired weight vector of the regression hyperplane shown in eq 8

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) \langle x_i, x \rangle + b \quad (8)$$

The last regression function is as follows

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) K(x_i, x_j) + b \quad (9)$$

$$K(x_i, x_j) = \exp(-\gamma \|x_i, x_j\|), \gamma > 0 \quad (10)$$

The values of the penalty parameter (C), the insensitive loss function (ε), and the kernel parameter (γ) will interact on the veracity of training data and predict data in the SVM model. The changes in the value of one parameter will cause changes of other parameters. The variable C represents the trade-off between model complexity of the training data and the model extension term with the mathematical meaning. The larger C value will result in overfitting of the training data set and minimization of the empirical risk without taking into account the more complex learning. However, the smaller C value will lead to the underfitting of the training data. The variable ε influences not only the complexity of the approximation function but also the number of support vectors.

2.1.2. Gray Wolf Optimizer. Mirjalili (2014)³⁷ developed a novel GWO optimization algorithm, which applied a new swarm intelligence algorithm inspired by gray wolves according to simulating the hierarchical relationship and hunting behavior of gray wolves spontaneously. With the advantage of other meta-heuristic algorithms such as genetic algorithm (GA),³⁸ ant colony optimization,³⁹ and particle swarm optimization,⁴⁰ the evolution process of these algorithm populations involves random factors, whereas the GWO algorithm represents better results compared to those of famous metaheuristics. The GWO algorithm mainly consists of four parts: social hierarchy, tracking, encircling, and attacking the prey process. In social hierarchy, GWO simulates the wolf hierarchies including four types of gray wolves: alpha (α), beta (β), delta (δ), and omega (ω), as shown in Figure 1. Gray wolves are quite gregarious in the apex predators of the food chain.

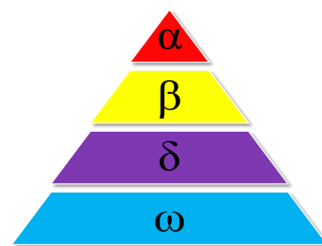


Figure 1. Gray wolf hierarchy (dominance decrease from top to down).

The most suitable solution is supposed as α . Analogously, β represents second-class solutions, δ represents third-class solutions, and ω represents the last remaining best solutions. In the progress of algorithm, α , β , and δ wolves lead the hunting, respectively, and the ω wolves follow.

At the mention of encircling prey, the behaviors could be mathematically modeled as follows

$$\vec{D} = |\vec{C} \cdot \vec{X}_p(t) - \vec{X}(t)| \quad (11)$$

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D} \quad (12)$$

where $\vec{X}$ represents the present position vector of a gray wolf, $\vec{X}_p$ is the present prey position, and $\vec{A}$ and $\vec{C}$ are, respectively, the coefficient vectors and emphasize exploitation and exploration obtained by

$$\vec{A} = 2\vec{a} \cdot \vec{r}_1 - \vec{a} \quad (13)$$

$$\vec{C} = 2 \cdot \vec{r}_2 \quad (14)$$

where $\vec{a}$ linearly descends from 2 to 0 with increasing iterations and $\vec{r}_1$ and $\vec{r}_2$ are, respectively, the random vectors from 0 to 1. Thus, $\vec{A}$ is a random value from $-a$ to a . For the sake of simulating the attacking prey, in the case of random values of $\vec{A}$ located in $[-1, +1]$, if $|\vec{A}| < 1$, the wolves must attack the prey and substitute for the present position of the prey.

Gray wolves guided by the α wolf in general can search and hunt for prey. We can get the α , β , and δ wolves in all iterations and enforce the other search agents to update their position with the consideration of current first-three class positions. The process of this portion is shown as

$$\vec{D}_i = |\vec{C}_j \cdot \vec{X}_j - \vec{X}| (i = \alpha, \beta, \delta; j = 1, 2, 3) \quad (15)$$

$$\vec{X}_j = \vec{X}_i - \vec{A}_j \cdot \vec{D}_i (i = \alpha, \beta, \delta; j = 1, 2, 3) \quad (16)$$

$$\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (17)$$

To summarize, the GWO algorithm randomly generates a set of gray wolves. According to the calculation of corresponding fitnesses and the possible position of the prey with the best three solutions, we can get the α , β , and δ wolves. Each candidate solution updates its position by eq 17. When hunting a prey, the attacking behaviors and emphasized exploration are under control of vectors $\vec{A}$ and $\vec{C}$. Finally, the GWO algorithm will be terminated once the last criterion is satisfied.

2.1.3. Adjusting the SVM Parameters Using the GWO Algorithm. In summary, correct hyperparameters (C , ε , and γ) have a great impact on the accuracy of predicted values by SVM. Therefore, we proposed the metaheuristic optimization methods, GWO-SVM model, looking for the proper SVM parameters in this work. The proposed novel GWO metaheuristic algorithm is supposed to have a great performance in the complex searching spaces (Table 1).³⁷ The flowchart of the GWO-SVM model is

Table 1. Parameters of the GWO-SVM Model to Predict the WDT

parameter	WDT
input data form	$[-1, +1]$
max iteration	100
search agent	30
best C	35.1846
best ε	0.0019
best γ	0.2231

shown in Figure 2 and parameters in Table 2. The GWO method is superior compared with traditional methods in the global search of complex spaces. Furthermore, the GWO-SVM model is good to deal with small sample data and shows a better performance in comparison with other machine learning algorithms such as ANN.^{41,42}

2.2. ANFIS. **2.2.1. Adaptive Network-Based Fuzzy Inference System.** By means of unifying the artificial neural network (ANN) and fuzzy inference system (FIS), ANFIS, representing the “adaptive network-based fuzzy inference system”, was modeled by Jang (1993)⁴³ to work out handling with the shortcomings of ANN and FIS. ANFIS as a fuzzy logic model running under rule-based approaches has been developed with its rules during the training procedure of the framework. ANFIS is a

five-layer system, and its layers are connected to each other. In this system, layer 0 implies the inputs and layer 5 represents the output. Between the two input and output layers, the hidden layers include numerous adjustable and fixed nodes. These nodes operate as the rules and membership functions (MFs). The ANFIS model contains x_1 , x_2 as two input variables and f as the output variable. The association of input and output nodes in the ANFIS framework is specified using “if then” fuzzy rules. Then, the two main fuzzy rules that rely on Takagi and Sugeno’s type are formulated as below

Rule 1

if x_1 is A1 and x_2 is B1 and etc. Then f1
 $= p1 \cdot x1 + q1 \cdot x2 + \dots + r1$

Rule 2

if x_2 is A2 and x_2 is B2 and etc. Then f1
 $= p2 \cdot x1 + q2 \cdot x2 + \dots + r2$

where $p1$, $q1$, $r1$ and $p2$, $q2$, $r2$ are the subsequent parameters; A1, B1, A2, and B2 are the arbitrary labels. Each layer is ascertained in the following:

Layer 1: Fuzzification layer

$$Q_{1,i} = \mu_{A_i}(x) \quad (18)$$

where each node i in layer 1 is an adaptive node with a node function as follows:

$Q_{1,i}$ is the output of the i th node; μ_{A_i} is a membership function (MF) like Gaussian MF, described as follows

$$\mu_{A_i}(x) = \exp \left[- \left(\left(\frac{x - c_i}{a_i} \right)^2 \right)^{b_i} \right], i = 1, 2 \quad (19)$$

where a , b , and c are the assumed parameters, which change the shape of the MF.

Layer 2: Rule layer

Every node in layer 2 is a constant node considered as Π ; besides, in the rule layer, the node output is the production of the arriving signals, such as

$$Q_{2,i} = W_i = \mu_{A_i}(x) \cdot \mu_{B_i}(x) \quad (20)$$

where $Q_{2,i}$ and W_i are the outputs of the rule layer and the firing strength of the i th rule, respectively.

Layer 3: Normalization layer

The W_i (firing strength) is normalized by the i th node in the third layer. Additionally, the ratio of the firing strength of the i th rule to the summation of firing strength of whole rules can be determined based on eq 21

$$Q_{3,i} = \overline{W}_i = \frac{W_i}{W_2}, i = 1, 2 \quad (21)$$

where $Q_{3,i}$ and W_i are the output of the third layer and the normalized firing strength, respectively.

Layer 4: Defuzzification layer

The node in the fourth layer is an adaptive node coupled with a node function, which represents the involvement of the i th rule with the general output. The output of fourth layer can be reckoned as follows

$$Q_{4,i} = \overline{W}_i f_i = \overline{W}_i (p_i \cdot x + q_i \cdot y + r_i), i = 1, 2 \quad (22)$$

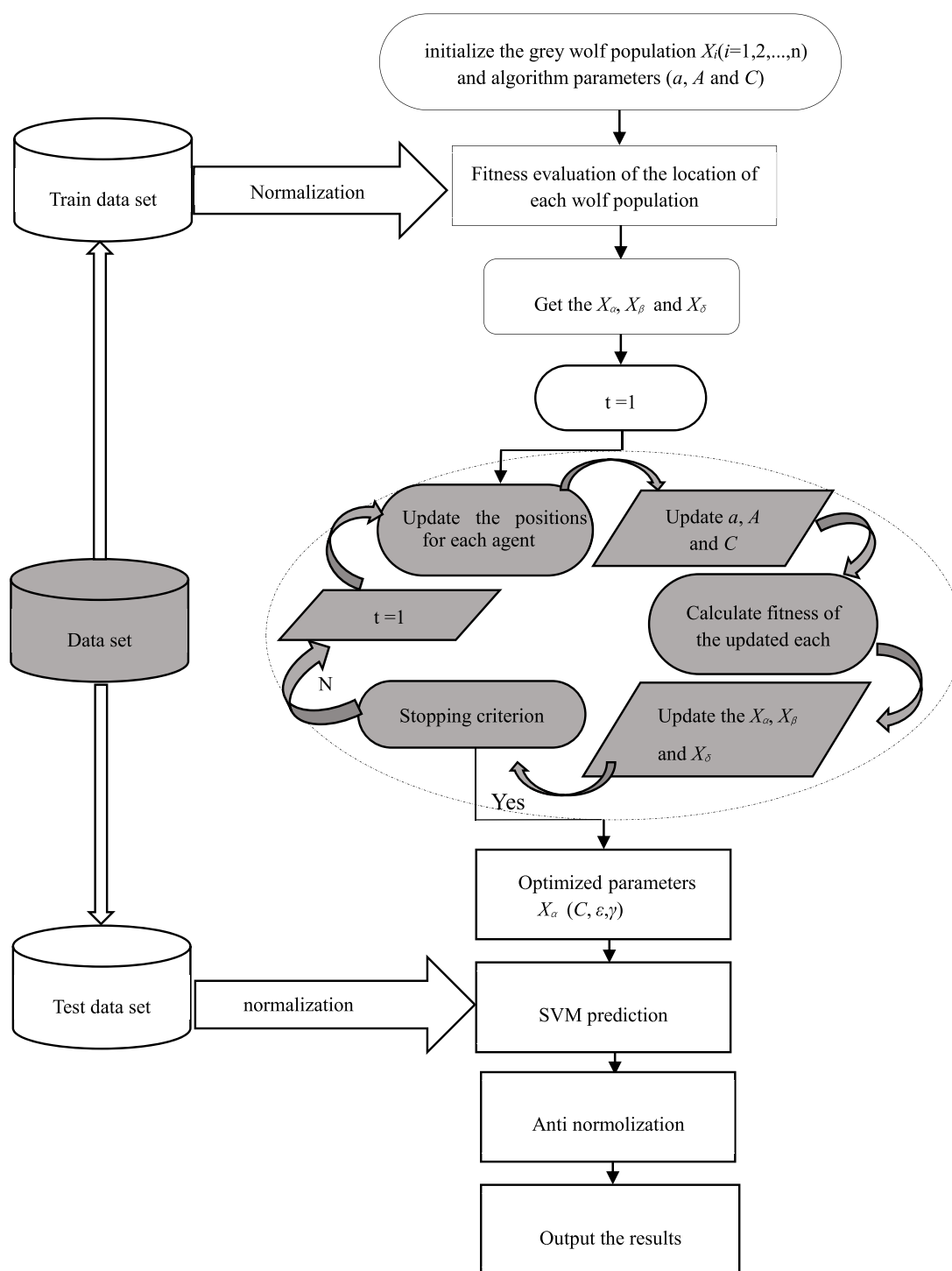


Figure 2. Flowchart of the newly proposed GWO-SVM model.

Layer 5: Output layer

To conclude, for calculating the output, we can employ eq 23 in layer 5

$$Q_{5,i} = \sum_i \bar{W}_i^f = \frac{\sum_i W_i^f}{\sum_i W_i}, i = 1, 2 \quad (23)$$

To define the MF parameters, the ANFIS algorithm can be utilized by a hybrid learning algorithm (Figure 3).

2.2.2. Particle Swarm Optimization (Induction Motor Control Using PSO-ANFIS). Particle swarm optimization^{44–46}

is able to optimize an objective function by applying a population-based searching. The population is made up of potential solutions, called particles, which are a metaphor of birds in flocks. These particles are randomly initialized and crossover the multidimension search space with freedom. During flight, each particle updates its own velocity and position on the basis of its own best experience and the whole population. The numerous procedures with the particle swarm optimization algorithm⁴⁵ are as follows:

Step 1: Randomly setting the velocity and position of all particles within predefined ranges.

Table 2. Summary of the Database of WDT Used in This Work

author	pressure (MPa)	composition	date	ref
M. Milhet (2004)	0.101–100	C14 + C15, C15 + C16	144	8
H.-Y. Ji (2004)	0.101	C14 + C15 + C16, C18 + C19 + C20, C13 + C16 + C17, C15/C17 + C19, C13 + C16/C17, C16 + C18/C20	74	20
J. L. Daridon (2002)	0.101–100	C13 – C24	54	7

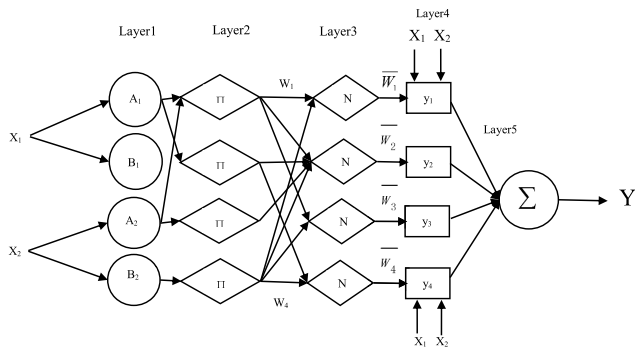


Figure 3. ANFIS architecture of two inputs, four rules, and first-order Sugeno model.

Step 2: Updating velocity: for each iteration, the velocities of every particle are updated with

$$v_i = v_i + c_1 \cdot R_1 \cdot (p_{i,best} - p_i) + c_2 \cdot R_2 \cdot (g_{i,best} - p_i) \quad (24)$$

where p_i and v_i are the position and velocity of particle i , respectively; $p_{i,best}$ and $g_{i,best}$ are the positions with the “best” objective value found so far by particle i and the entire population, respectively. w is a parameter controlling the behaviors of flying; R_1 and R_2 vary randomly in the range of $[0,1]$; c_1 and c_2 are the factors controlling the correlation weighting of corresponding terms. The random variables are beneficial to the stochastic searching of PSO.

Step 3: Updating position: the positions of all particles are updated with

$$p_i = p_i + v_i \quad (25)$$

p_i will be checked and limited in the allowed range after updating

Step 4: Updating memory: $p_{i,best}$ and $g_{i,best}$ are updated when satisfied

$$p_{i,best} = p_i, \text{ if } f(p_i) > f(p_{i,best}) \quad (26)$$

$$g_{i,best} = g_i, \text{ if } f(g_i) > f(g_{i,best}) \quad (27)$$

where $f(x)$ is optimized as the objective function

Step 5: Stopping: The algorithm duplicates from steps 2 to 4 until exact stopping situations are satisfied, such as a predefined number of iterations. When stopped, the algorithm gets the values of g_{best} and $f(g_{best})$ as its solution.

PSO⁴⁷ applies many searching points approaching the global optimal point by degrees with its p_{best} and g_{best} . Original positions of p_{best} and g_{best} are different. However, applying the different directions of p_{best} and g_{best} , all agents gradually approach the global optimum.

2.2.3. Genetic Algorithm. In the first step, it should be described that evolution can be defined as an optimization process. Hence, just like the other intelligent methods, the origin of evolution can be observed in nature. The evolutionary calculating data was carried out regarding computer instructions of the algorithm and evolutionary process.⁴⁸ Weismann’s and Darwin’s theory (1984)⁴⁹ and genetics of Mendel are the accepted evolution

theory as a set of arguments known.⁵⁰ The essential characteristics of genetic algorithm (GA) can look for another region in the specific area that we can hint at “detecting many routs arbitrarily and instantaneously” as one of them.⁵¹ The initial and main operators of GA are crossover, mutation, and inversion.⁵² The balance of exploitation of the forming process with searching for new regions is persisted by the initial operators. To operate GA, a primary population is created with consideration of a problem. Then, the objective function considers the uniqueness of each solution process for problems. Hereafter, the initial population is put in order relevant to the fitness of every individual. Taking into account that crossover and mutation operators on the correct and appropriate individual continue till the fulfillment of final criteria, the next generation can be constructed.

2.3. LSSVM. The LSSVM model is a vastly improved and well-organized version of SVM. LSSVM is an appropriate model for a data set containing numerous values due to equation linearization. Also, this model is more implicit and has a better convergence rate. In other words, this method is able to unravel difficult problems by finding the quickest way while the converging level is optimum. At present, the structure of the optimized network in the training stage has been entirely developed, which leads to attaining an optimum hyperplane with a minimum error to target data.⁵³ To clarify the differences between SVM and LSSVM, both methods are described. Assume that $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$, $x_i \in R^N$, $y_i \in R^N$. n and N are the input data and output data, the training data sample with a size equal to the training data and the number of input variables. Therefore, input variable x is of dimension $N \times n$. Hence, the regression (eq 28) relying on SVM fundamental terms can be described as follows

$$y_k = w^T \varphi(x_k) + b, w \in R^N \rightarrow R^M, M \rightarrow \infty \quad (28)$$

where $\varphi(x_k)$, b , w , and w^T are the nonlinear kernel function mapping the input data to a multidimensional space, the bias term, the weight factor, and the transposed weight matrix. The following cost function was proposed by Vapnik,⁵⁴ which can be minimized to reckoned the bias and weight factors

$$\text{cost } t - \text{function} = \frac{1}{2} w^T + c \sum_{k=1}^n (\zeta_k - \zeta_k^*) \quad (29)$$

which must meet the constraints mentioned as follows

$$\begin{aligned} y_k - w^T \varphi(x_k) - b &\leq \varepsilon + \zeta_k, k = 1, \dots, n \\ w^T \varphi(x_k) + b - y_k &\leq \varepsilon + \zeta_k^*, k = 1, \dots, n \\ \zeta_k, \zeta_k^* &\geq 0, k = 1, \dots, n \end{aligned} \quad (30)$$

where y_k and x_k explain the output of the k th data point. Also, ε and ζ_k, ζ_k^* are exactness of the function estimation and slack variables, respectively. To prevent unachievable solution for proposing a highly precise and accurate model, the value of ε should not be too small because some values may not be in the range of the ε precision. Therefore, the slack variables are employed for having a safe error bound. Moreover, the coefficient c (>0) in eq 32, as a yardstick of deviation degree from an

earlier defined precision ε , explains an optimizing factor in the SVM model. Equation 31 introduces the proposed cost function eq 30, which is minimized via the Lagrangian for these problematic terms, eq 4 is correlated as follows

$$L(\alpha, \alpha^*) = -\frac{k}{2} \sum_{k=1}^n k(\alpha_k - \alpha_k^*)(x_k - x_1) - \varepsilon \sum_{k=1}^n (\alpha_k - \alpha_k^*) + \sum_{k=1}^n y_k(\alpha_k - \alpha_k^*) \quad (31)$$

$$\sum_{k=1}^n (\alpha_k - \alpha_k^*) = 0, \alpha_k, \alpha_k^* \in [0, c] \quad (32)$$

where the α_k and α_k^* are the Lagrangian multiplier parameters. Taking into account the previous modifications, the final description of SVM can be defined as the following correlation

$$K(x_k, x_1) = \varphi(x_k)^T \varphi(x_1), k = 1, \dots, n \quad (33)$$

where α_k , α_k^* , and b parameters can be calculated by utilizing quadratic programming to unravel the nonlinear problem. Subsequently, the least-squares modification of SVM (LSSVM) has been emerged to prevent such problematic complexity, which means LSSVM fits a linear correlation between dependent and independent variables. In other words, the fittest correlation can be defined as the one that reduces the cost function to the lowest level with a regression error term (e_k). Moreover, the SVM correlation can be reformed as follows⁵⁴

$$\cos t\text{-function} = \frac{1}{2} w^T w + \frac{1}{2} \gamma \sum_{k=1}^n e_k^2 \quad (34)$$

where the linear constraint is as follows

$$y_k = w^T \varphi(x_k) + b + e_k \quad (35)$$

As it is mentioned in eq 37, the first term is a classic L_2 norm on the regression weights; however, the burden of representing the regression error term (e_k) for n training data sets is in the hand of the second term, where the regularization constant ($\gamma \geq 0$) is the LSSVM parameter for tuning. For optimization, this parameter can be helpful because it represents the comparative weight of the second and the first term. The regression error expresses the deviation of estimated data from experimental values, which can be defined under the name of constraint. The precision of the LSSVM model is relevant to error definitions. Regarding the fact that the regression error in the SVM model is considered in a different way, all of those errors that are smaller than the fixed value are neglected. The LSSVM model involves the equality completely different from SVM, which deals with inequality constraints. Hence, to achieve the ideal final relation, the linear constraint as mentioned in the cost function (eq 35) can be used, whereby, instead of unraveling a nonlinear and quadratic problem, we can solve a set of linear correlations as a desirable solution. For optimizing the problematic solutions, the Lagrange function is correlated as

$$L(w, b, e, \alpha) = \frac{1}{2} w^T w + \frac{1}{2} \gamma \sum_{k=1}^n e_k^2 - \sum_{k=1}^n \alpha_k (w^T \varphi(x_k) + b + e_k - y_k) \quad (36)$$

With Lagrangian multipliers $\alpha_k \in \mathbb{R}$, according to the Lagrangian multipliers, the subsequent conditions are assumed to achieve the optimal cost function

$$\frac{\partial L}{\partial w} = 0 \rightarrow w = \sum_{k=1}^n \alpha_k \varphi(x_k)$$

$$\frac{\partial L}{\partial b} = 0 \rightarrow \sum_{k=1}^n \alpha_k = 0$$

$$\frac{\partial L}{\partial e_k} = 0 \rightarrow \alpha_k = \gamma e_k, k = 1, \dots, n$$

$$\frac{\partial L}{\partial \alpha_k} = 0 \rightarrow w^T \varphi(x_k) + e_k + b - y_k = 0, k = 1, \dots, n \quad (37)$$

The organized system of equations using the entire data set involves $2N + 2$ equations and $2N + 2$ unknowns such as α_k , e_k , w , and b . Having solved the system containing all equations, it was ascertained that the LSSVM parameters like γ as the tuning parameter are gained.

As mentioned earlier, SVM and LSSVM approaches are utilized to model both linear and nonlinear correlations; hence, they deal with the kernel function and their variables should be optimized by the use.⁴³ Any function that affirms Mercer's assumptions or conditions can be referred as a kernel function. Two vectors of x and x_k in the $\varphi(x)$ and $\varphi(x_k)$ spaces are the producers of kernel function ($K(x, x_k)$) as their inner product that leads to $K(x, x_k) = \varphi(x)^T \varphi(x_k)$. Kernel functions remove the requirement of straightly calculating the values ($\varphi(x)$), while involving the feature spaces of any random or desired dimension. Eventually, the RBF kernel as defined under the name of the radial basis function can be utilized⁵⁵ and is described as eq 38

$$K(x, x_k) = \exp\left(-\frac{\|x_k - x\|^2}{\sigma^2}\right) \quad (38)$$

In eq 38, σ^2 , as another tuning parameter of the kernel function, is relevant to optimized LSSVM model. Overall, a LSSVM model provided with a RBF kernel function contains γ and σ^2 as two tuning parameters. Minimizing the difference of the LSSVM framework from the actual data, tuning parameters can be measured.⁵⁴

3. DATA ANALYSIS

3.1. Experimental Data. To develop and evaluate novel models, 272 data sets of experimental WDT collected from the existing literature^{7,8,20} were used in this work. The obtained

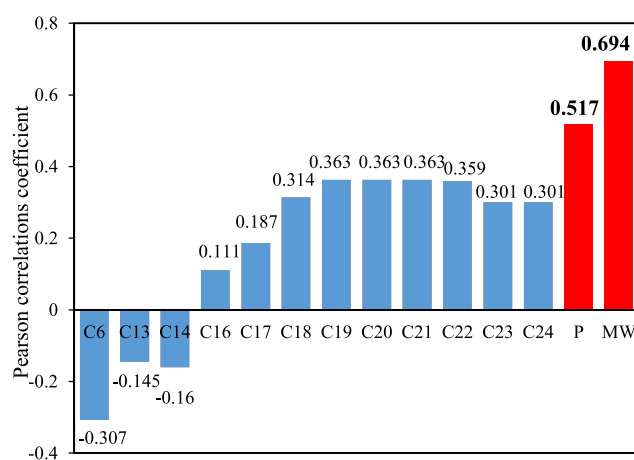


Figure 4. Pearson correlations for WDT.

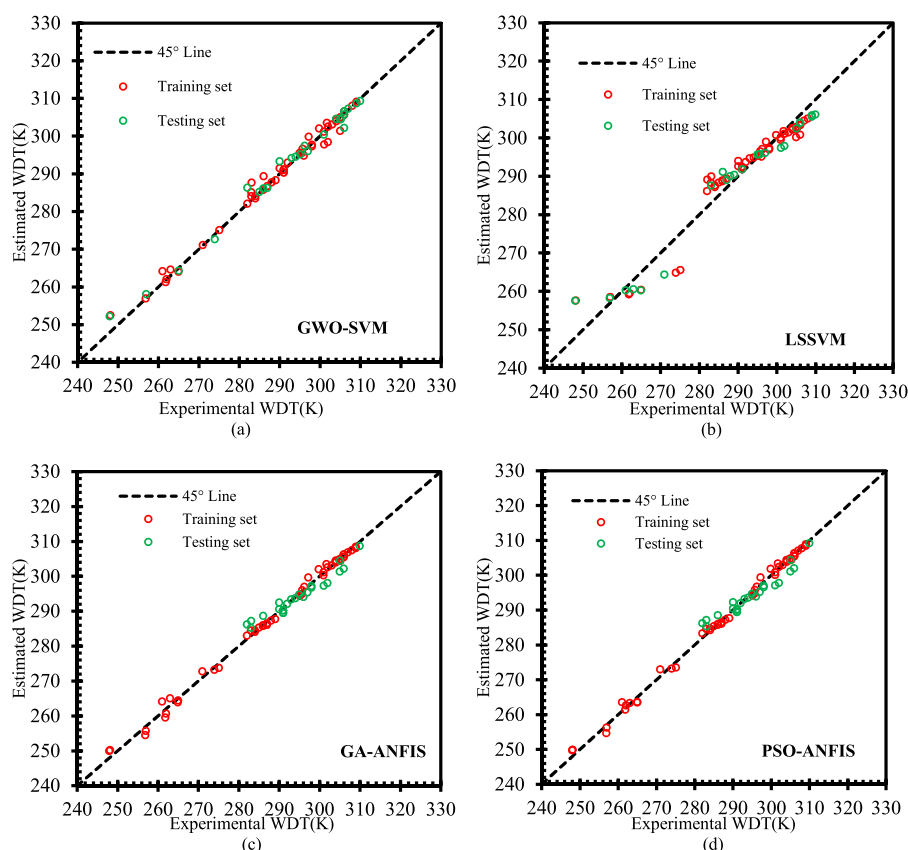


Figure 5. (a–d) Estimated WDT versus experimental WDT by the proposed models with a ternary system (74 data sets).

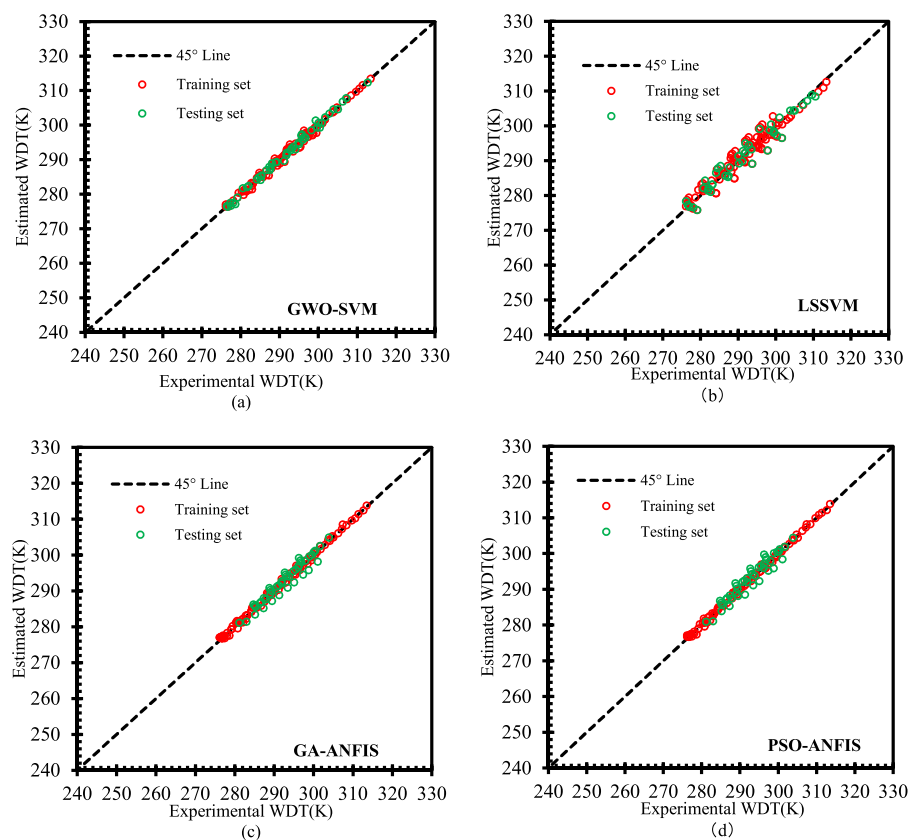


Figure 6. (a–d) Estimated WDT versus experimental WDT by the proposed models with a binary system (144 data sets).

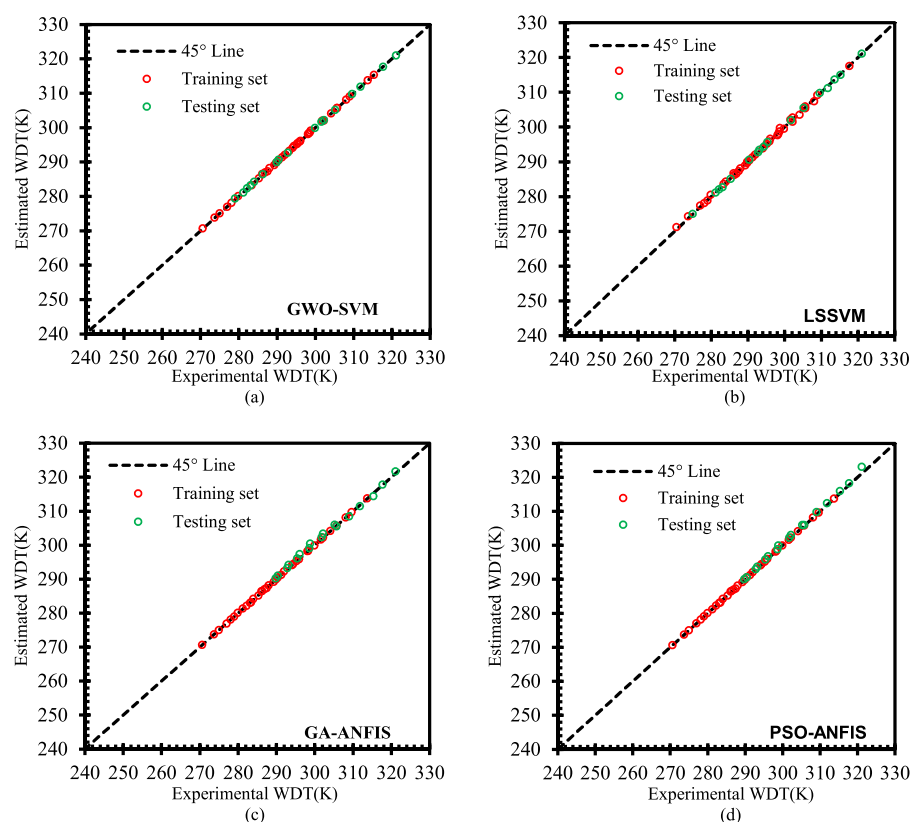


Figure 7. (a–d) Estimated WDT versus experimental WDT by the proposed models with a multicomponent system (54 data sets).

Table 3. Comparison of Four Models To Predict WDT with a Ternary System (74 Data Sets)

evaluation		GWO-SVM	GA-ANFIS	PSO-ANFIS	LSSVM
training set	AARD/%	0.3633	0.3261	0.6303	0.9282
	SD	0.0058	0.0044	0.0039	0.0197
	RMSE	1.6122	1.1770	2.2994	3.5927
	R^2	0.9877	0.9955	0.9964	0.9282
	EP _{max} /%	1.7283	1.1771	0.9670	3.8162
testing set	EP _{min} /%	0.0066	0.0037	0.0033	0.0264
	AARD/%	0.4130	0.6303	0.6491	1.0465
	SD	0.0064	0.0080	0.0082	0.0137
	RMSE	1.7332	2.2994	2.3689	3.6516
	R^2	0.9879	0.8412	0.8329	0.9545
total	EP _{max} /%	1.6957	1.4573	1.4749	3.8702
	EP _{min} /%	0.0040	0.0182	0.0187	0.0379
	AARD/%	0.3794	0.4288	0.4097	1.0046
	SD	0.0059	0.0058	0.0056	0.0132
	RMSE	1.6524	1.6338	1.6060	3.6112
	R^2	0.9878	0.9585	0.9388	0.9403
	EP _{max} /%	1.7284	1.4574	1.4750	3.8703
	EP _{min} /%	0.0040	0.0038	0.0033	0.0264

experimental data sets covered pressures from 0.1 to 100 MPa and number of carbon atoms from 6 to 24. The data sets were randomly divided into two parts, with training sets comprising 70% of all data points and testing sets comprising 30% of all data points. The training sets were applied for regulating the parameters (C , ϵ , γ) with an objective to minimize the average mean-square error. Data sets from Milhet⁸ are composed of two binary systems, i.e., including C14 + C15 and C14 + C16 at pressures from 0.1 to 100 MPa. The data from Ji²⁰ are almost composed of various ternary systems (just a very few data sets are composed of a binary system) at atmospheric pressure only.

And the data sets from Daridon⁷ are made up of a regular distribution of n -paraffins ranging from C13 to C24 with various numbers of components at pressures from 0.1 to 100 MPa (details in Table 2). For the sake of testing and validating those four models' performances about binary, ternary, and multi-component systems at atmospheric pressure only and in a wide range of pressures of 0.1–100 MPa, respectively, we predict three different data sets in this work.

3.2. Influential Factors. The main factors influencing WDT of paraffin are pressure, composition, carbon atom number, and molar mass.

Table 4. Comparison of the Four Models To Predict WDT with a Binary System (144 Data Sets)

evaluation		GWO-SVM	GA-ANFIS	PSO-ANFIS	LSSVM
training set	AARD/%	0.1928	0.1465	0.1362	0.5972
	SD	0.0027	0.0019	0.0018	0.0111
	RMSE	0.7870	0.5508	0.5038	2.1300
	R ²	0.994	0.9326	0.9178	0.9489
	EP _{max} /%	0.6978	1.7383	0.6502	0.6882
testing set	EP _{min} /%	0.0011	0.0042	0.0009	0.0016
	AARD/%	0.1785	0.3985	0.4585	0.5728
	SD	0.0024	0.0052	0.0057	0.0073
	RMSE	0.7060	1.5032	1.6435	2.1201
	R ²	0.9940	0.9326	0.9178	0.9486
total	EP _{max} /%	0.6628	1.7383	1.0131	1.1493
	EP _{min} /%	0.0052	0.0495	0.0486	0.0110
	AARD/%	0.1878	0.2235	0.2347	0.5897
	SD	0.0025	0.0033	0.0034	0.0073
	RMSE	0.7599	0.9493	1.0008	2.1298
	R ²	0.9927	0.9886	0.9874	0.9378
	EP _{max} /%	0.6978	1.7383	1.0131	1.1493
	EP _{min} /%	0.0011	0.0042	0.0009	0.0016

Table 5. Comparison of the Four Models To Predict WDT with a Multicomponent System (54 Data Sets)

evaluation		GWO-SVM	GA-ANFIS	PSO-ANFIS	LSSVM
training set	AARD/%	0.0311	0.0312	0.0257	0.1125
	SD	0	0	0	0.0014
	RMSE	0.1240	0.1147	0.1075	0.4007
	R ²	0.9999	0.9999	0.9999	0.9985
	EP _{max} /%	0.1113	0.3038	0.2285	0.1836
testing set	EP _{min} /%	0.0041	0.0142	0.0012	0.0001
	AARD/%	0.0419	0.2237	0.1858	0.0651
	SD	0	0.0027	0.0024	0
	RMSE	0.1433	0.7747	0.7142	0.2560
	R ²	1	0.9927	0.9945	0.9996
total	EP _{max} /%	0.0617	0.1141	0.5328	0.6083
	EP _{min} /%	0.0155	0.0022	0.0237	0.0570
	AARD/%	0.0345	0.0918	0.761	0.0976
	SD	0	0.0015	0.0013	0.0012
	RMSE	0.1304	0.4449	0.4105	0.3615
	R ²	0.9999	0.9986	0.9988	0.9990
	EP _{max} /%	0.1113	0.3038	0.5328	0.6083
	EP _{min} /%	0.0041	0.0022	0.0012	0.0001

On the one side, we use the Pearson correlation analysis to explore the influence of the major factors on WDT. The bigger the value of the Pearson correlation coefficient between any input and output variable, the more significant the influence of the input variable on determining the value of the output variable. The result of Pearson correlation analysis is shown in Figure 4, where it was realized that the sensitive parameters on the WDT are, in the ascending order, C15, C16, C13, C14, C17, C18, C24, C23, C6, C19, C22, C20, C21, pressure, and molar mass. On the other side, it is well-known that the WDT will increase with the increasing pressure, molar mass, composition, and carbon atom number from the existing literature. Furthermore, the influence of molar mass and pressure on WDT is greater than that of other factors. Therefore, the proposed four models were built to get the best performance and regression between molar mass and pressure as input and WDT as output.

4. RESULTS AND DISCUSSION

To validate the performance of the developed models, a comparison among the GWO-SVM, LSSVM, GA-ANFIS, and PSO-ANFIS is

made for all 272 data sets, which contrasts data of experimental and estimated WDT by models shown in Figures 5–8. As illustrated in Figures 5–8, the data sets estimated by the proposed GWO-SVM gathered closer with the 45° line than those by other three intelligence algorithm models, proving that values predicted by GWO-SVM have an excellent agreement with experimental WDT.

Many assessment parameters, i.e., the average absolute relative deviation (AARD), root-mean-squared error (RMSE), standard deviation (SD), determination coefficient (R²), and error percentage (EP), were applied as a further method to evaluate different models, whose formulas for these assessment parameters are shown as follows

$$\text{AARD} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i^{\text{exp}} - y_i^{\text{cal}}}{y_i^{\text{exp}}} \right| \quad (39)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i^{\text{exp}} - y_i^{\text{cal}})^2} \quad (40)$$

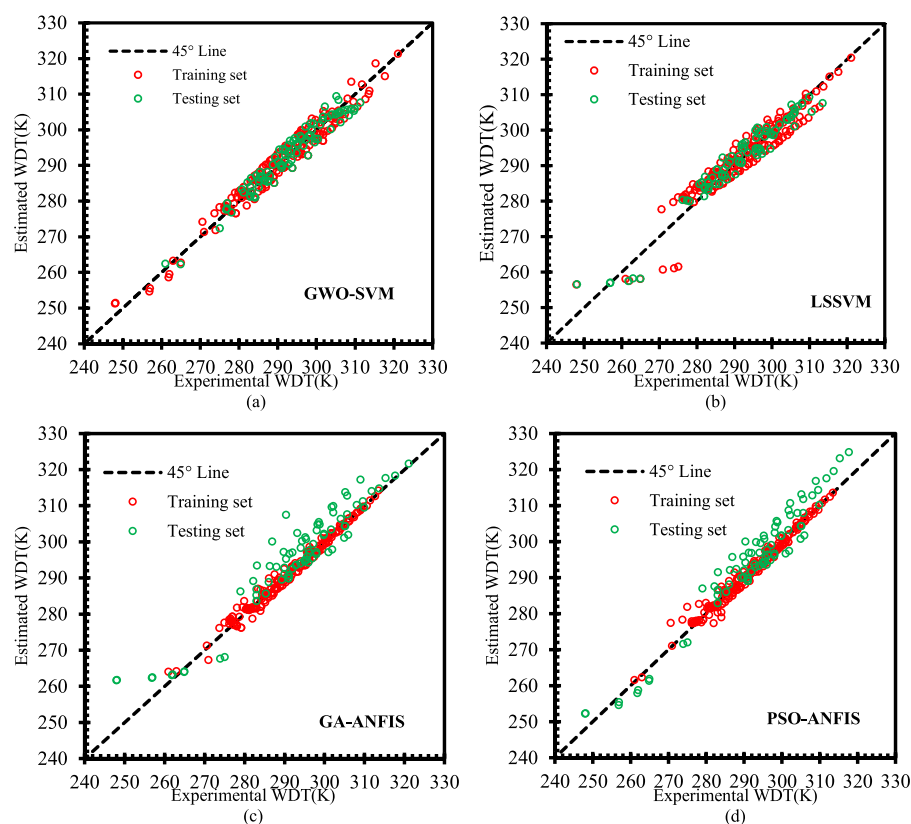


Figure 8. (a–d) Estimated WDT versus experimental WDT by the proposed models with all 272 data sets.

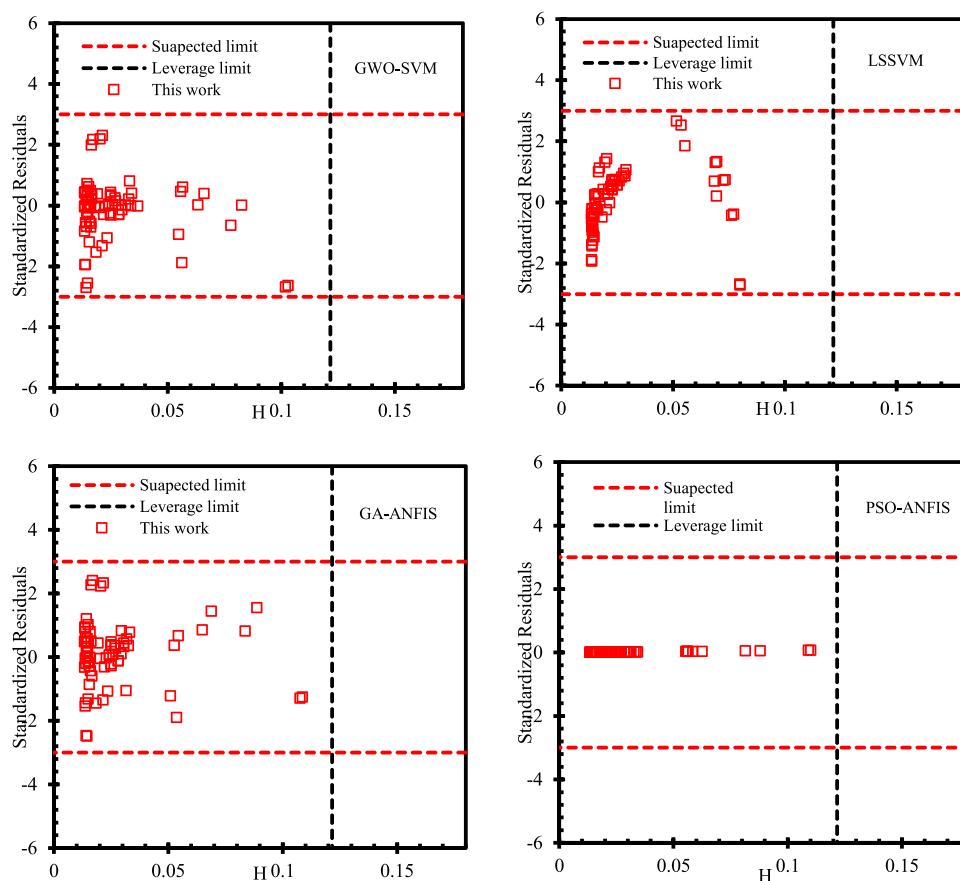


Figure 9. Detection of doubtful data in the four models using hat values with a ternary system (74 data sets).

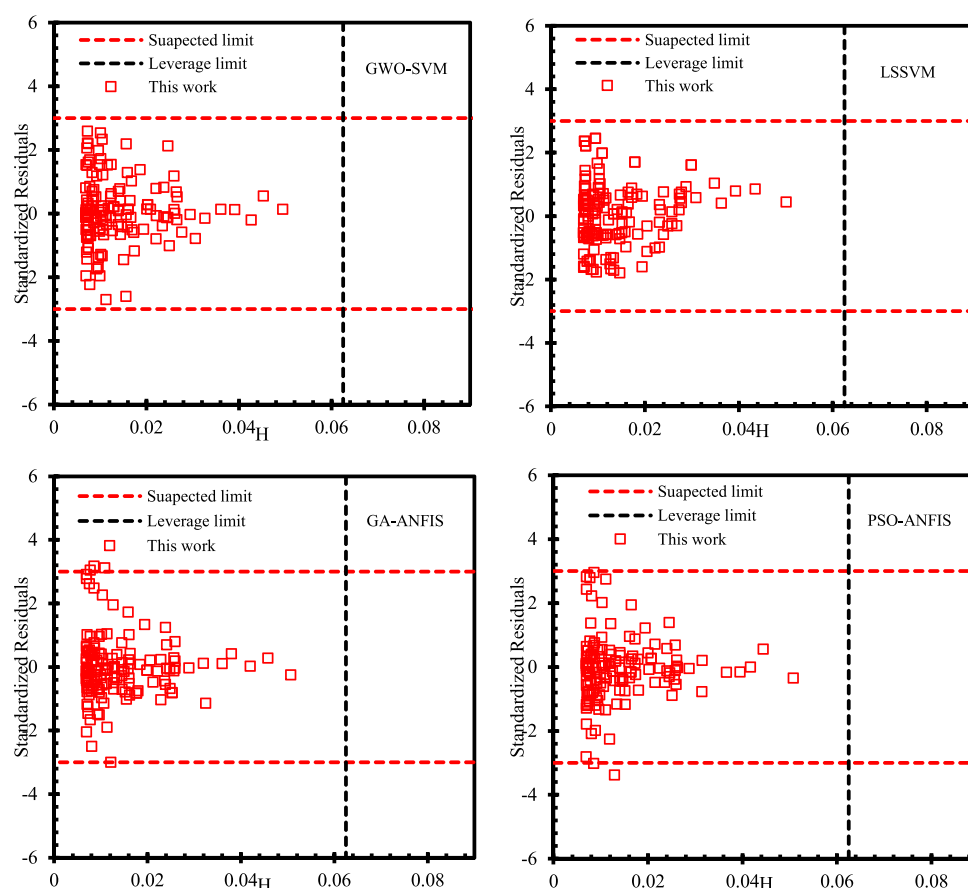


Figure 10. Detection of doubtful data in the four models using hat values with a binary system (144 data sets).

$$SD = \sqrt{\frac{1}{N-1} \sum_{i=1}^N \left(\frac{y_i^{\text{exp}} - y_i^{\text{cal}}}{y_i^{\text{exp}}} \right)^2} \quad (41)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i^{\text{exp}} - y_i^{\text{cal}})^2}{\sum_{i=1}^N (y_i^{\text{exp}} - y_{\text{ave}}^{\text{cal}})^2} \quad (42)$$

$$EP = 100 \times \frac{|y_i^{\text{exp}} - y_i^{\text{cal}}|}{y_i^{\text{exp}}} \quad (43)$$

where N is the amount of all experimental WDT data points; y_i^{cal} and y_i^{exp} are calculated and experimental WDTs, respectively, and $y_{\text{ave}}^{\text{cal}}$ represents the average value of all experimental WDTs.

The comparison results of the WDT among the proposed GWO-SVM, GA-ANFIS, PSO-ANFIS, and LSSVM models are given in Tables 3–6.

It is seen from Table 6 and Figure 8 that the proposed GWO-SVM model has the best performance in the four models in this work and provides predictions in extremely excellent agreement with all experimental WDTs with AARD of 0.7128% and R^2 of 0.9546 in comparison with the other three proposed models (GA-ANFIS, PSO-ANFIS, and LSSVM) with AARD of 0.7483, 0.7518, and 0.8818% and R^2 of 0.9291, 0.9474, and 0.9127, respectively. Thus, it is known that GWO-SVM shows the best agreement in all 272 data sets (value of AARD is minimum and R^2 is maximum in four models). To deeply evaluate the developed models, a comparison of predicted values of the models with the experimental data of different ternary (74 data sets) (2004),²⁰

binary (144 data sets) (2004),⁸ and multicomponent (54 data sets) (2002)⁷ systems is respectively illustrated in Figures 5–7 and Tables 3–5. Figures 5–7 show a good agreement that can be observed between the estimated and experimental WDTs (closely scattered points around the line of 45°) of models of different systems. Tables 3–5 indicate certain values about the performance of four models. The AARD of the ternary system of GWO-SVM is 0.3794, lower than that of other models, where AARD values for GA-ANFIS, PSO-ANFIS, and LSSVM are 0.4288, 0.4097, 1.0046, respectively from Table 3. The R^2 of the ternary system of GWO-SVM is 0.9878, higher than that of other models, where R^2 values for GA-ANFIS, PSO-ANFIS, and LSSVM are 0.9585, 0.9388, and 0.9403, respectively, from Table 3. The AARD of the binary system of GWO-SVM is 0.1878, lower than that of other model, where AARD values for GA-ANFIS, PSO-ANFIS, and LSSVM are 0.2235, 0.2347, and 0.5897, respectively, from Table 4. The R^2 of the binary system of GWO-SVM is 0.9987, higher than that of other model, where R^2 values for GA-ANFIS, PSO-ANFIS, and LSSVM are 0.9886, 0.9874, and 0.9378, respectively, from Table 4. The AARD of the multicomponent system of GWO-SVM is 0.0345, lower than that of other model, where AARD values for GA-ANFIS, PSO-ANFIS, and LSSVM are 0.0918, 0.761, and 0.0976, respectively, from Table 5. The R^2 of the multicomponent system of GWO-SVM is 0.9999, higher than that of other model, where R^2 values for GA-ANFIS, PSO-ANFIS, and LSSVM are 0.9986, 0.9988, and 0.9990, respectively, from Table 5. Thus, it is been known that GWO-SVM is the best controller, which presented satisfactory performances no matter which system it is in.

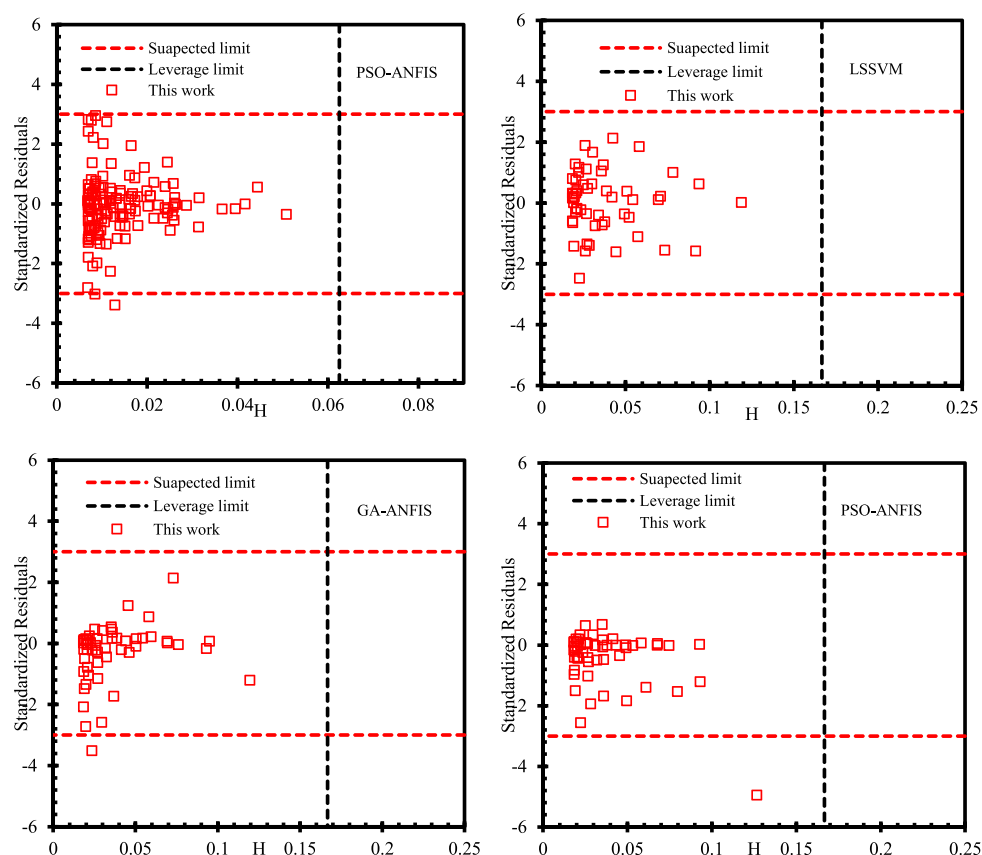


Figure 11. Detection of doubtful data in the four models using hat values with a multicomponent system (54 data sets).

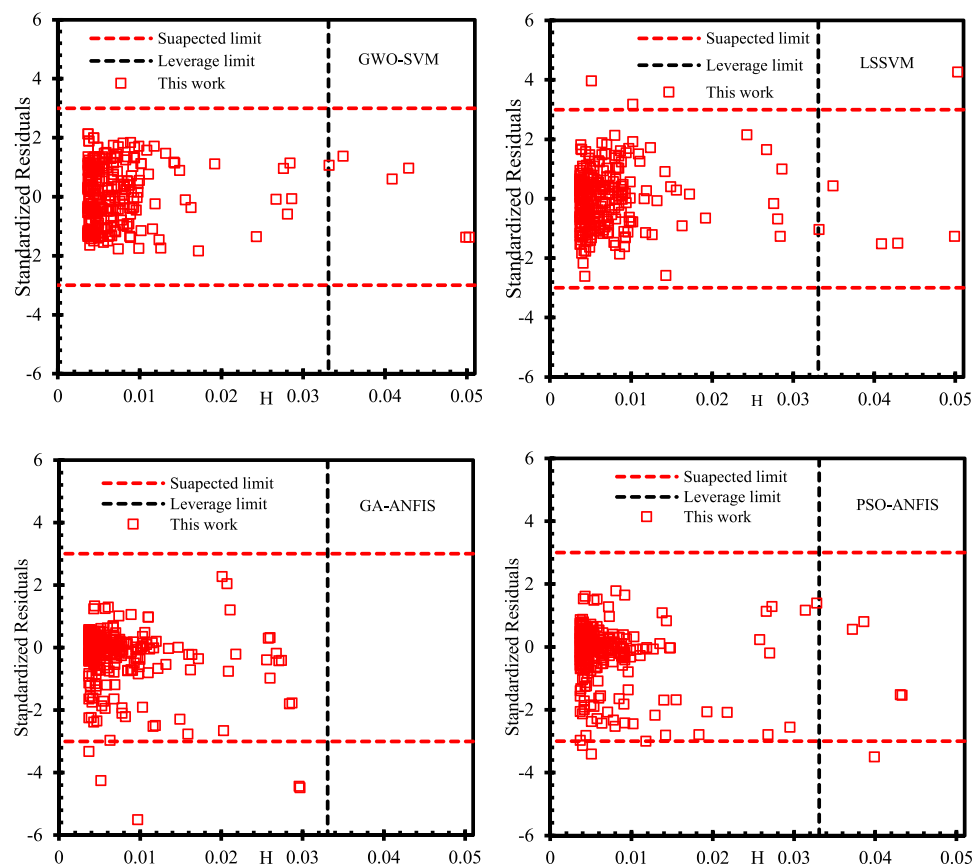


Figure 12. Detection of doubtful data in the four models using hat values with all 272 data sets.

Table 6. Comparison of the All 272 Data Sets

evaluation		GWO-SVM	GA-ANFIS	PSO-ANFIS	LSSVM
training set	AARD/%	0.7303	0.3855	0.3509	0.9313
	SD	0.0079	0.0047	0.0055	0.0184
	RMSE	2.4289	1.5542	1.3256	3.4173
	R^2	0.9592	0.9822	0.9751	0.8934
	EP _{max} /%	1.7279	4.9341	2.6180	2.8546
testing set	EP _{min} /%	0.0191	0.0010	0.0033	0.0011
	AARD/%	0.6922	1.2688	1.3375	0.7672
	SD	0.0091	0.0188	0.0155	0.0100
	RMSE	2.4038	4.5378	5.2682	2.8235
	R^2	0.9426	0.8704	0.9233	0.9460
total	EP _{max} /%	1.7279	3.3819	5.8797	3.3075
	EP _{min} /%	0.0178	0.0025	0.0070	0.0281
	AARD/%	0.7128	0.7483	0.7518	0.8818
	SD	0.0083	0.0110	0.0096	0.0115
	RMSE	2.4208	3.0975	2.8098	3.2498
	R^2	0.9546	0.9291	0.9474	0.9127
	EP _{max} /%	1.7279	3.3819	5.8797	3.3075
	EP _{min} /%	0.0178	0.0010	0.0033	0.0011

The leverage method was applied for searching the outlier data points in the data set for stability analysis. For the sake of detection of doubtful data sets, Williams' plot is sketched by determining the hat values, as shown in Figures 9–12.

From Figures 9, 10 and 1, 2, it is known that there is no doubtful data in 54, 74, 144 data sets in GWO-SVM and LSSVM, although there are several data sets in other two models. Moreover, there are 6 doubtful data points in GWO-SVM, which are less than those in LSSVM (8 doubtful data points). Thus, GWO-SVM is not only effective in the statistics field but also had stability and strong robustness in reflecting the intrinsic relationship, and it exhibits the highest performance statistically in all proposed models.

5. CONCLUSIONS

Four metaheuristic models (GWO-SVM, LSSVM, GA-ANFIS, and PSO-ANFIS) were proposed to determine the accurate value of WDT. According to Pearson correlation analysis, reservoir pressure and molar mass are the major factors for WDT. All data sets collected from the existing literature were divided into two parts, of which 70% of all data points were trained and 30% of all data points were tested. Besides, the presented GWO-SVM model was compared with other three metaheuristic models (LSSVM, GA-ANFIS, and PSO-ANFIS). It turns out that under any condition, i.e., binary or ternary systems and single pressure of 0.1 MPa or a range of pressures from 0.1 to 100 MPa, the GWO-SVM model provides the closest agreement with experimental data with certain values assessing models being mentioned in Tables 3–6 and closely scattered points around the 45° line in Figures 5–8. Besides, GWO-SVM has the characteristics of great stability and robustness ascertained via Williams' plot applied to verify these characteristics. Furthermore, GWO-SVM has a better performance in the multicomponent system than in the binary and ternary systems.

This work presents an efficient framework for estimating WDT by combining GWO and SVM together. As the core of the framework, GWO integrates the distinctive inherent sensitivity, ergodicity, and randomness of theory, which is able to explore much more searching space, therefore selecting the most relevant features for prediction of WDT. However, there are only 272 data points that are not enough to cover different systems, for example,

for some systems containing more numbers of carbon atoms. As a result, models will get an unsatisfied performance. Also, more comprehensive and profound inequality constraints are needed for SVM.

As a consequence, the proposed method will provide a practical way for technical support in petroleum engineering and has a profound impact on the development of crude oil and gas condensate. It will give a favor for WDT of high-pressure condensate reservoir to develop difficulty.

In future works, the other mechanisms such as orthogonal learning or quadratic interpolation will be investigated to further enhance the search capability of GWO. In addition, the GWO can be fused into the computer-aided system for much easier decision-making.

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Notes

The authors declare no competing financial interest.

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